

Daily Review



1 Name the three energy stores.

Chemical, grav.
potential, elastic

2 What does a battery store?

Chemical energy

3 A wind turbine's input energy?

Kinetic (of the wind)

4 A toaster: one wasted form?

Light (or sound)

5 Where does wasted energy go?

Spreads out as heat

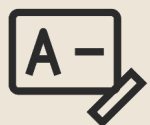
Efficiency, Power & Teams

How much energy does the job — and how fast?

S3 Engineering Science



What the pictures mean



Show me boards
Hold up your board.



Write it
Write it in your jotter.



Turn and talk
Talk to your partner.



Hands down
No hands up. I pick.



Hands up
Hand up to answer.



Silent task
Work alone, in silence.



Jotter notes
Copy it into your jotter.



Cut and stick
Cut out and stick in.

Councils are swapping every street light for LEDs. Why is that worth the cost?



LED street lighting

PART 1

Efficiency, Power and Teams

Calculate it, explain why it matters, and work like an engineer.

By the end of this part...



Rate each one 1-3: 1 not yet, 2 nearly, 3 got it.

- 1 I can calculate percentage efficiency
- 2 I can calculate power using $P = E \div t$
- 3 I can explain how energy use affects cost and the environment
- 4 I can describe the marble run's energy changes
- 5 I can describe good engineering teamwork, using what + why

SECTION A

Efficiency

An old bulb is 10 % efficient. Where does the other 90 % go?

Efficiency – How Much Does the Job?

Formula

efficiency = useful output ÷ total input × 100

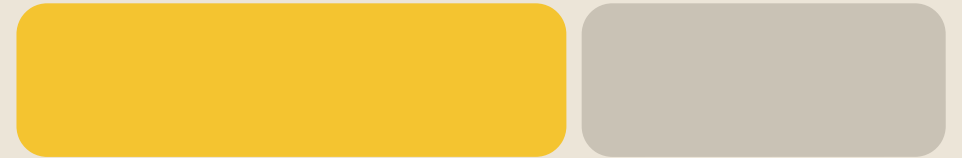
Always

a percentage (%)

Never

more than 100 %

100 J in



60 J useful

40 J

wasted

$$60 \div 100 \times 100 = 60 \%$$

Worked Example – An LED Lamp

An LED lamp takes in 50 J and gives out 45 J of light. Calculate its efficiency.

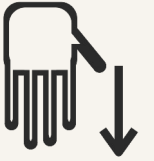
Relationship $\text{efficiency} = \text{useful} \div \text{input} \times 100$

Substitute $= 45 \div 50 \times 100$ (1)

Answer + unit $= 90 \%$ (1)

1 mark for the numbers in. 1 mark for the answer with %.

We Do — An Electric Drill



A drill takes in 200 J and 130 J does useful work. Calculate its efficiency.

Relationship $\text{efficiency} = \text{useful} \div \text{input} \times 100$

Substitute $= 130 \div 200 \times 100$

Answer + unit $= 65 \%$

Faded — A Radio



A radio takes in 400 J and gives out 100 J of sound. Calculate its efficiency.

Relationship $\text{efficiency} = \text{useful} \div \text{input} \times 100$

Substitute $= 100 \div 400 \times 100$

Answer + unit $= 25 \%$

Why Engineers Improve Efficiency

Trap: “to waste less energy” scores 0.

Cost

less energy needed, so lower bills

Environment

less fuel burned, so less CO₂

Cooler

less heat, so it lasts longer



Show Me — Efficiency



1

Toaster: 1000 J in, 600 J useful.
Efficiency?

60 %

2

Scooter: 2500 J in, 2250 J useful.
Efficiency?

90 %

3

Boiler: 5000 J in, 3500 J useful.
Efficiency?

70 %

Show Me — Efficiency



4

Someone gets 125 %. What went wrong?

Divided the wrong way
round

5

Why improve efficiency? What + why.

Less energy — so lower
bills

Show Me — Efficiency

EXTRA PRACTICE



1 3000 J in, 900 J useful. Efficiency?

30 %

2 Old bulb: 60 J in, 6 J of light.

10 %

3 1000 J in, 650 J useful. How much is wasted?

350 J

Show Me — Efficiency

EXTRA PRACTICE



4

A is 45 %, B is 70 %. Which wastes less?

B — so it costs less

5

One way engineers improve efficiency.

Insulation, less friction, sensors

SECTION B

Power, Cost and the Planet

Two kettles boil the same water. One is faster. Why?

Power — How Fast Energy Is Transferred

$$P = E \div t$$

P

power in watts
(W)

E

energy in joules
(J)

t

time in seconds
(s)

1 watt = 1 joule transferred every second.

Worked Example – A Toaster

A toaster transfers 36 000 J of energy in 30 s. Calculate its power.

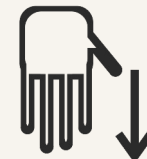
Relationship $P = E \div t$

Substitute $= 36\,000 \div 30$ (1)

Answer + unit $= 1200\text{ W}$ (1)

P power (W) E energy (J) t time (s)

We Do — A Hairdryer



A hairdryer transfers 45 000 J in 30 s. Calculate its power.

Relationship $P = E \div t$

Substitute $= 45\,000 \div 30$

Answer + unit $= 1500\text{ W}$

P power (W) E energy (J) t time (s)

Faded — A Fridge



A fridge transfers 6000 J in 60 s. Calculate its power.

Relationship $P = E \div t$

Substitute $= 6000 \div 60$

Answer + unit $= 100 \text{ W}$

P power (W) E energy (J) t time (s)

Energy Use → Cost and the Environment

Cut both: switch off, sensors, insulation, efficient kit.

Cost

More energy used



More electricity bought



A bigger bill

Environment

More electricity generated



More gas burned



More CO₂ released

Show Me — Power



1 Kettle: 90 000 J in 30 s. Power?

3000 W

2 Lamp: 600 J in 60 s. Power?

10 W

3 Phone charger: 1200 J in 60 s. Power?

20 W

4 The unit of power?

Watts (W)

5 Lights left on all night: effect on cost?

More energy — so a
bigger bill

Show Me — Power

EXTRA PRACTICE



1 Vacuum cleaner: 42 000 J in 30 s.

1400 W

2 Games console: 9000 J in 60 s.

150 W

3 A: 600 J in 10 s. B: 600 J in 20 s.
More powerful?

A — same energy, less
time

Show Me — Power

EXTRA PRACTICE



4

TV on standby all day: effect on the environment?

More generated — so
more CO₂

5

One way a school could cut its energy bill.

Sensor lights,
insulation

SECTION C

The Marble Run and Teams

You can't add energy to a marble run. So where does it all go?

The Marble Run – The Energy Story

1 Start

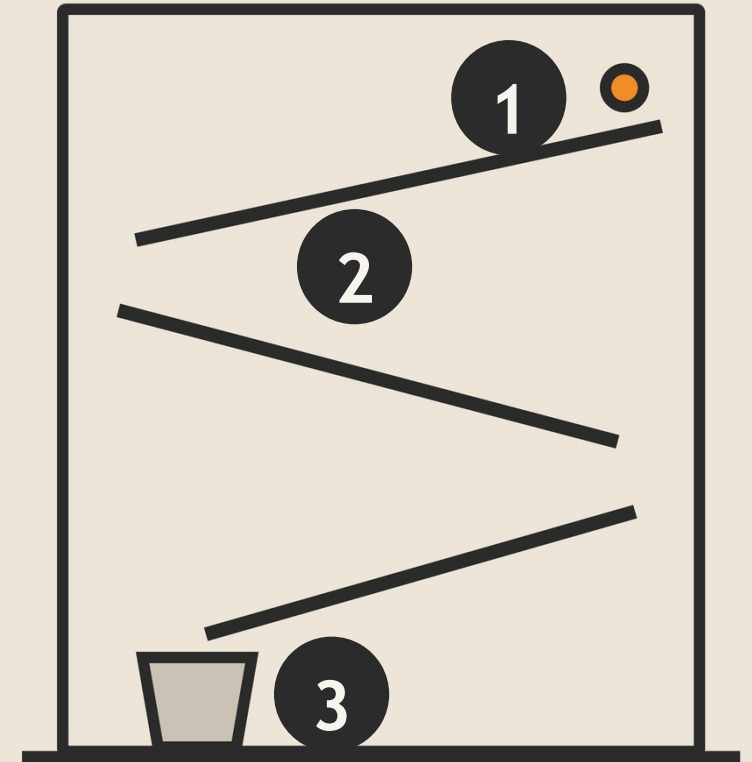
gravitational potential – it's high up

2 Rolling down

gravitational potential $\rightarrow$ kinetic

3 All the way

friction: some energy $\rightarrow$ heat and sound, wasted



Worked Example – A Bouncing Ball

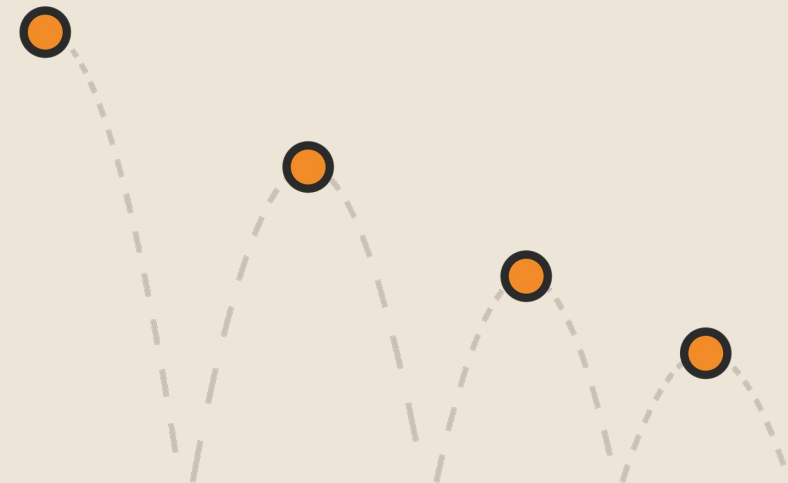
Explain why a ball doesn't bounce back to where it was dropped from.

What

each bounce transfers some energy to heat and sound

Why

less kinetic energy is left, so it can't climb as high



Good Teamwork — What + Why

Trap: “we worked together” scores 0.

Share jobs

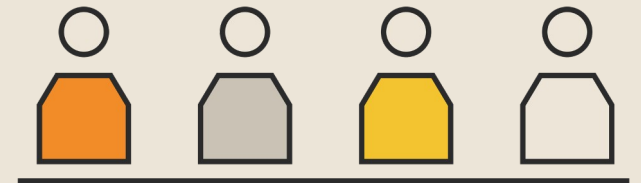
so everyone has one and it gets done

Listen

so the best idea is picked

Agree a sketch

so everyone builds the same design



Show Me — Marble Run and Teams



1 Energy at the top of the run?

Gravitational potential

2 Energy when it's fastest?

Kinetic

3 Two forms wasted on a run?

Heat and sound

4 Why can nothing be higher than the start?

It can't gain energy

5 One thing a good team does, what + why.

Shares jobs — so all get
done

Show Me — Marble Run and Teams

EXTRA PRACTICE



1 Skateboarder at the top of a half-pipe?

Gravitational potential

2 ...at the bottom?

Kinetic

3 Why do her swings get smaller?

Friction → heat and
sound

Show Me — Marble Run and Teams

EXTRA PRACTICE



4

One member has nothing to do. Fix it.

Give a role, e.g.
Recorder

5

Why agree a sketch before building?

So all build the same
design

A pupil gets an efficiency of 125 %. What does that tell you?



- A a very good machine
- B it makes extra energy
- C they divided the wrong way round
- D nothing — it's fine

Answer C: Efficiency can't pass 100 %. Useful $\div$ input, not input $\div$ useful.

“Why do engineers improve efficiency?” Which answer scores?



- A “to waste less energy”
- B “to make it efficient”
- C “less electricity needed, so lower bills”
- D “it's better”

Answer C: A and B just say what efficiency is. C says why it matters.

Power is measured in...



- A joules
- B seconds
- C watts
- D percent

Answer C: Energy is in joules. Power — joules per second — is in watts.

Mini Test 4



Mini Test 4 sheet

On your own, in silence, pen and calculator.

12 minutes – 10 marks.

Then swap and mark from the next slides.

Mini Test 4 — Marking

ANSWERS

Q1 (1 + 1 marks)

$$1500 \div 5000 \times 100 (1) = 30 \% (1)$$

Q2 (1 mark)

Less energy is wasted as heat, so less heat builds up.

Q3 (1 + 1 marks)

$$P = 72\,000 \div 90 (1) = 800 \text{ W } (1). \text{ No unit: lose mark 2.}$$

Mini Test 4 – Marking

ANSWERS

Q4 (1 + 1 marks)

Cost: more electricity used, so a bigger bill. Environment: more generated, so more fuel burned, more CO₂.

Q5 (1 + 1 marks)

(a) gravitational potential → kinetic. (b) friction transfers kinetic energy to heat.

Scores 0

a bare “costs more” or “pollution”

Mini Test 4 — Marking

ANSWERS

Q6 (1 mark)

Give everyone a job or role (e.g. Recorder, tester), so the work is shared and finished in time.

Scores 0

a bare “work together”

Brain Dump



In your jotter

Write everything you remember about efficiency, power and teamwork.

On your own, 2 minutes. No notes.

Then we check it against the Plenary.

Plenary – Efficiency, Power, Teams



- 1 The efficiency formula?
- 2 The power formula and its units?
- 3 One reason to improve efficiency, what + why.

Useful $\div$ input $\times$ 100

$P = E \div t$ – W, J, s

Less energy – so lower
bills

Plenary – Efficiency, Power, Teams



4 The marble run's energy story?

Grav. potential →
kinetic

5 One thing a good team does, what +
why.

**Listens – so best idea
wins**

Plenary – Efficiency, Power, Teams

EXTRA PRACTICE



- 1 Can efficiency be over 100 %?
- 2 800 J in 20 s. Power?
- 3 Lights left on: one environmental effect?
- 4 Where does the marble's energy go?
- 5 300 J in, 120 J useful. Efficiency?

No

40 W

More CO₂ from power stations

Heat and sound
(friction)

40 %

Look back — by the end of this part...



Rate each one 1-3 again. Compare with the start.

- 1 I can calculate percentage efficiency
- 2 I can calculate power using $P = E \div t$
- 3 I can explain how energy use affects cost and the environment
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LOOKING AHEAD

Test 1 — Wednesday 30 September

30 minutes, 30 marks. The data booklet is allowed. Bring a pen.

Credits

The big question (LED street lighting): ©

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Why Engineers Improve Efficiency: © Roger A

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